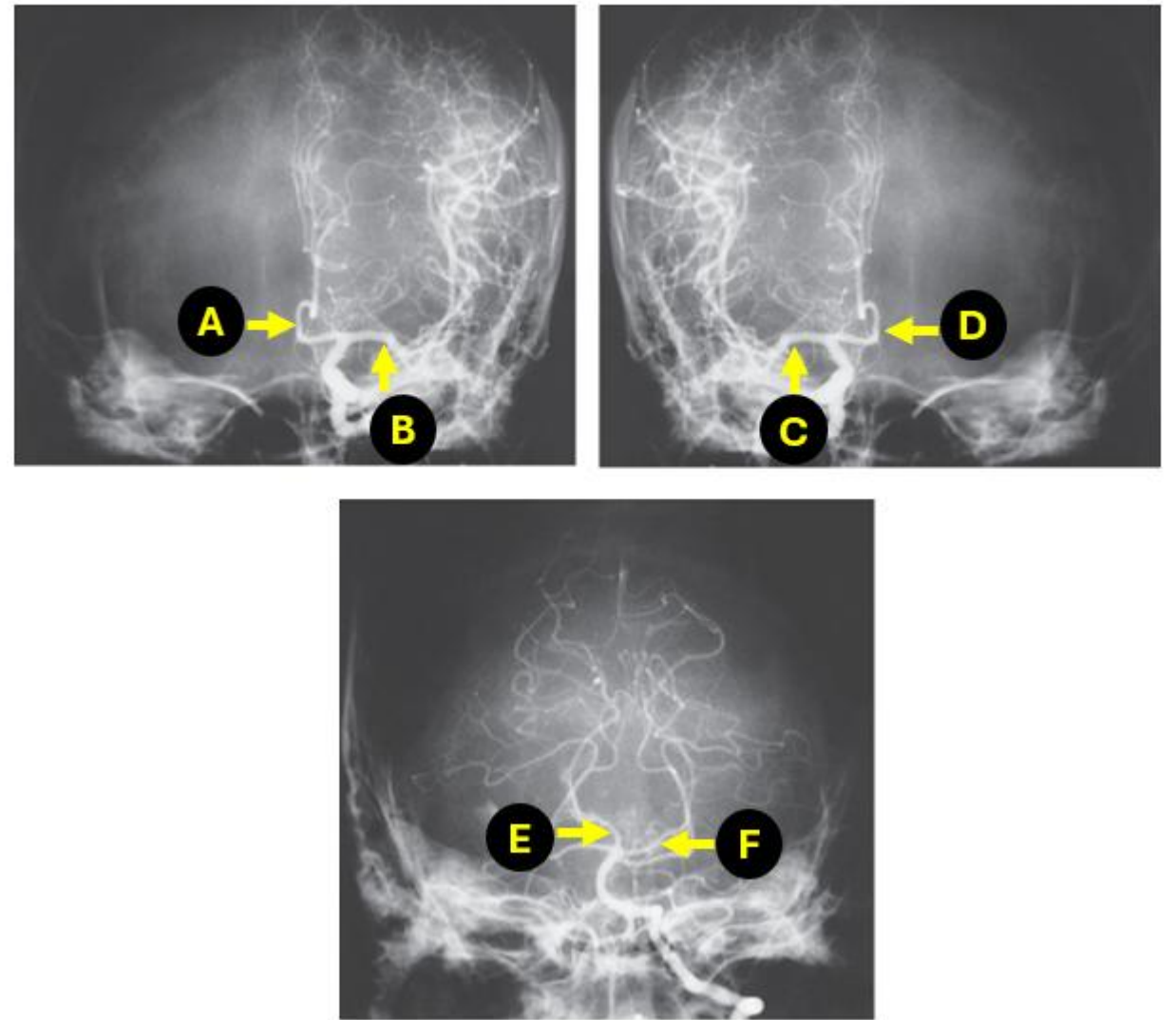


NB IMCQ 3

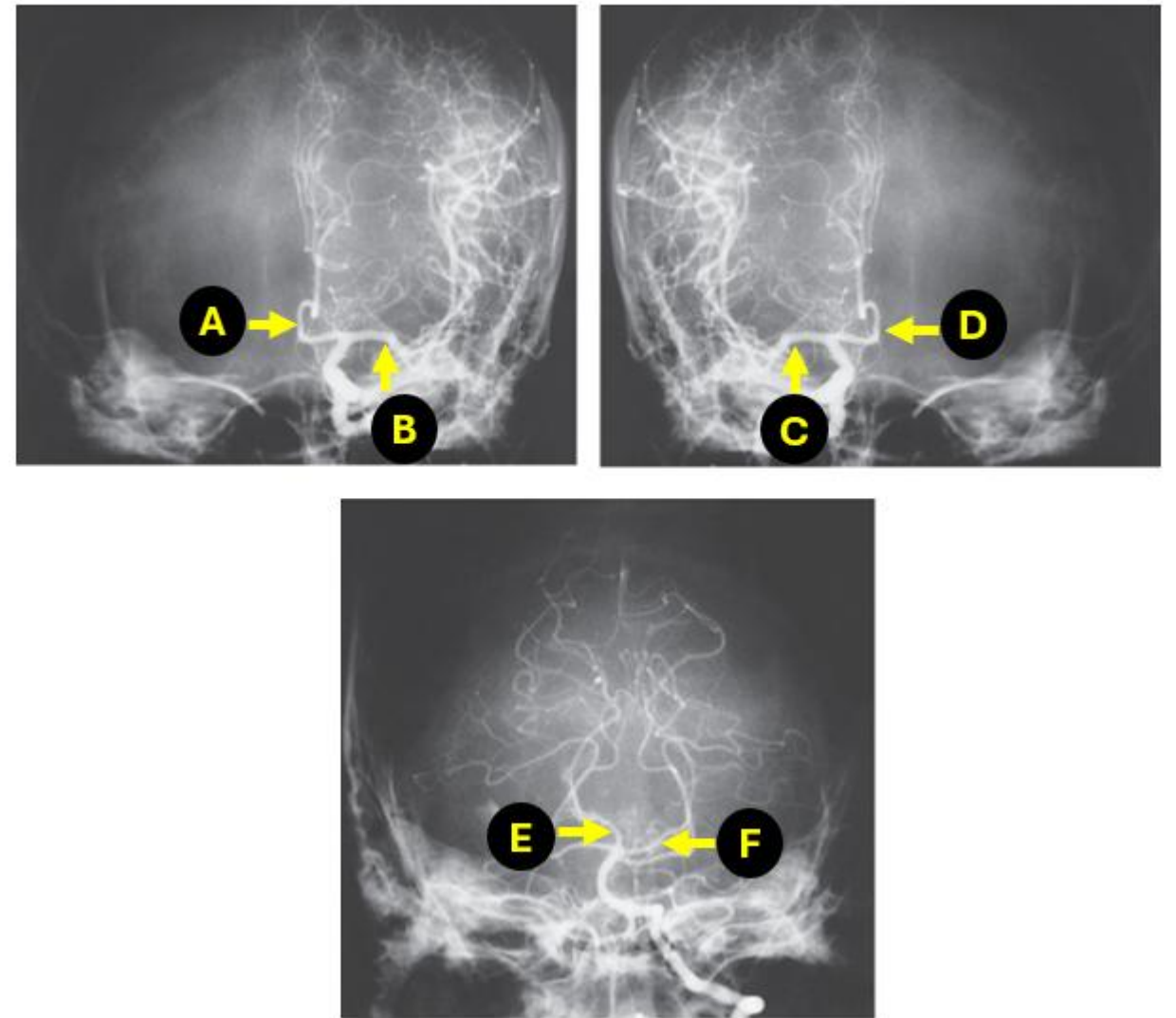
Neuroscience, Histology, Anatomy

1. A 47-year-old woman is brought to the emergency department because of the sudden onset of difficulty walking. Neurologic examination shows reduced muscle strength in the right lower extremity. Sensation to pinprick is decreased over the right lower extremity. Cranial nerves are intact. Neuroimaging shows a unilateral cerebral infarction. The most likely cause of the neurologic findings in this patient is occlusion of which of the following labeled arteries in the angiograms of a normal brain?

- A. A
- B. B
- C. C
- D. D
- E. E
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SOM.MK.I.BPM2.5.NB.5.NSCI.0360 Examine the subarachnoid arteries

SOM.MK.I.BPM2.5.NB.5.NSCI.0559 Relate the functional deficits experienced by a patient in a clinical scenario to the functional anatomy of the central nervous system

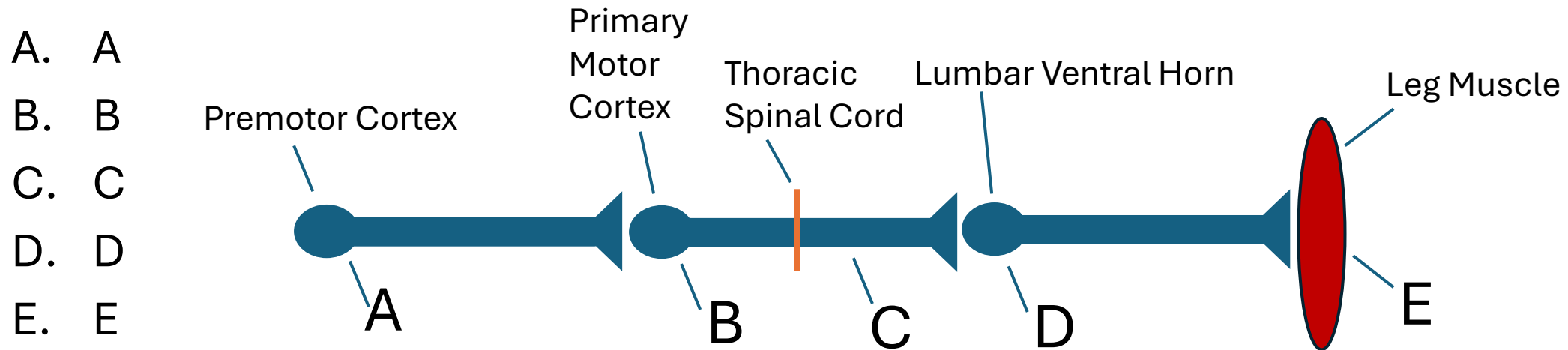
2. A 70-year-old woman is brought to the physician because of a 1-hour history of nausea and vomiting after she spilled an undiluted insecticide on her leg while working in her garden. Physical examination shows pallor and diaphoresis. During the examination, she develops a seizure associated with the release of high levels of an excitatory amino acid by central neurons. Which of the following glial cells is most likely responsible for the uptake of the implicated neurotransmitter in this patient?

- A. Astrocytes
- B. Ependymal cells
- C. Microglia
- D. Oligodendrocytes
- E. Schwann cells

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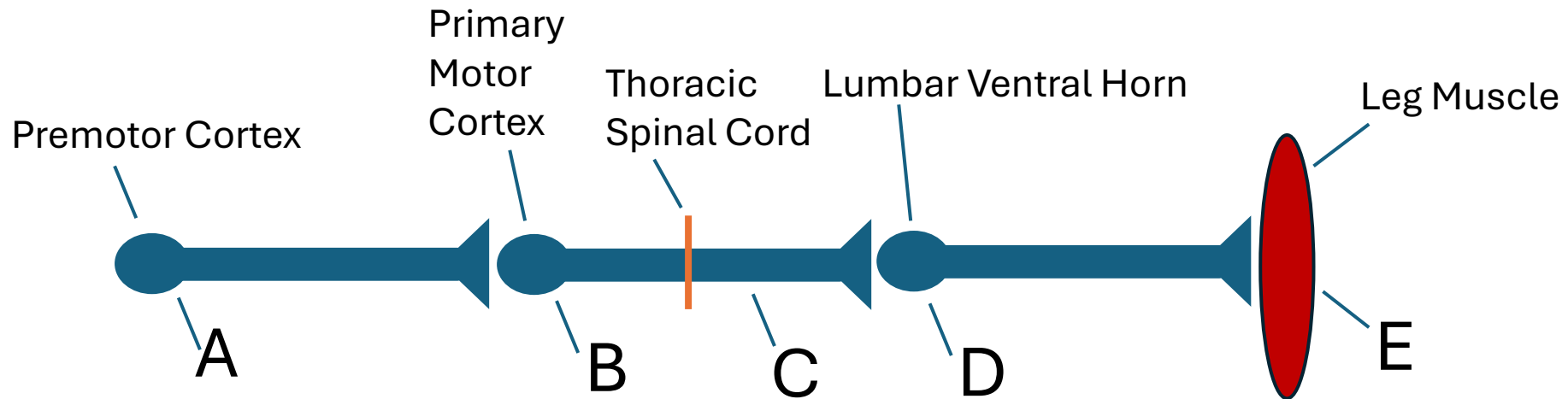
3. A 36-year-old man is brought to the emergency department because of a 2-hour history of immobility of his left leg after he sustained a stab wound to his back. Imaging studies show a spinal injury at left T8. It is expected that descending axons of upper motor neurons are transected at this level (see orange line), disrupting the flow of motor information from cortical motor areas to the muscles of the affected leg. Which of the following sites, indicated in the figure below, is most likely to undergo retrograde transneuronal degeneration in this patient?



3. A 36-year-old man is brought to the emergency department because of a 2-hour history of immobility of his left leg after he sustained a stab wound to his back. Imaging studies show a spinal injury at left T8. It is expected that descending axons of upper motor neurons are transected at this level (see orange line), disrupting the flow of motor information from cortical motor areas to the muscles of the affected leg. Which of the following sites, indicated in the figure below, is most likely to undergo retrograde transneuronal degeneration in this patient?



- A. A
- B. B
- C. C
- D. D
- E. E



4. A 52-year-old man is brought to the emergency department 1 hour after sustaining a puncture wound to the left arm. He says that his left thumb is numb and tingling. Neurologic examination shows decreased sensation to light touch over his left thumb. Damage to which of the following cell morphotypes best explains the sensory manifestations of the injury in this patient?

- A. Astrocytes
- B. Bipolar neurons
- C. Microglia
- D. Multipolar neurons
- E. Pseudo-unipolar neurons
- F. Oligodendrocytes

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Histology

5. A 48-year-old woman is brought to the emergency department 50 minutes after she sustained left facial injuries in a motor vehicle collision. Her vital signs are within normal limits. Physical examination shows a puncture wound of the left eye that is limited to the lateral part of the corneoscleral limbus. Which of the following is the most likely complication of injury to this ocular region?

- A. Inability to accommodate
- B. Inability to produce aqueous humor
- C. Loss of the transparency of the lens
- D. Damage of Bruch's membrane
- E. Vascularization of the cornea
- F. Retinal detachment
- G. Infection of the sebaceous glands of Zeis

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SOM.MK.I.BPM2.4.NB.1.HCB.1820 Give the function(s) of the cornea of the corneoscleral coat.

SOM.MK.I.BPM2.4.NB.1.HCB.1821 Describe the microscopic structures of the cornea of the corneoscleral coat

6. A 64-year-old man comes to the physician because of a 3-month history of dizziness. Physical examination shows aggravation of dizziness when his head is moved up and down in a linear fashion. A dysfunction of the vestibule of the inner ear is suspected. Which of the following histological features best characterizes the sensory region of the affected structure?

- A. Crista ampullaris
- B. Auditory ossicles
- C. Cupula
- D. Otolithic membrane
- E. Tympanic membrane
- F. Tectorial membrane
- G. Cortilymph
- H. Hair cells without a true kinocilium

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SOM.MK.I.BPM2.4.NB.1.HCB.1858 Describe the histological structure of the maculae of the saccule and utricle

Anatomy

7. A 64-year-old woman comes to the physician because of a 2-weeks history of difficulty opening her left eye and episodes of headaches that became increasingly worse over the past month. She has a history of hypertension. Physical examination shows the left pupil remains central when the patient is asked to look to the left side. The right pupil moves normally. Imaging studies show an aneurysm of the carotid artery within the cavernous sinus. Which of the following cranial nerves is most likely being compressed?

- A. X
- B. IX
- C. VI
- D. V1
- E. IV

7. A 64-year-old woman comes to the physician because of a 2-weeks history of difficulty opening her left eye and episodes of headaches that became increasingly worse over the past month. She has a history of hypertension. Physical examination shows the left pupil remains central when the patient is asked to look to the left side. The right pupil moves normally. Imaging studies show an aneurysm of the carotid artery within the cavernous sinus. Which of the following cranial nerves is most likely being compressed?

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8. A 45-year-old man comes to the physician because of a 2-day history of swollen upper lip, headache and generalized facial pain. Physical examination shows a non-fluctuant abscess with diffused erythema on the upper lip. Imaging studies show cavernous sinus thrombosis. Which of the following additional findings is most likely in this patient?

- A. Loss of smell
- B. Loss of pupillary light reflex
- C. Loss of taste to anterior 2/3 of tongue
- D. Loss of sensation to the gingiva of lower mandible
- E. Dryness of the mouth

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SOM.MKII.BPM2.1.NB.1.ANAT.0242

Describe the cause of cavernous sinus syndrome, the nerves that may be affected and what consequences can occur

9. A 40-year-old man is brought to the emergency department because of a 3-day history of fever, neck pain, and difficulty swallowing. Physical examination shows erythema and swelling in the anterior neck with tenderness along the midline. CT scan of the neck shows an infection spreading between the buccopharyngeal fascia and the alar layer of the deep cervical fascia. If untreated, this infection is most likely spread into which of the following spaces?

- A. Carotid sheath
- B. Parapharyngeal space
- C. Prevertebral space
- D. Retropharyngeal space
- E. Suprasternal space

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SOM.MKII.BPM2.1.NB.1.ANAT.0064

Describe the fascial layers of the neck including investing fascia, pretracheal and buccopharyngeal fascia, prevertebral fascia and alar fascia and their clinical relevance

10. A 64-year-old man is brought to the emergency department because of severe respiratory distress following an upper airway obstruction. The physician performs an emergency incision through the cricothyroid membrane to establish an airway. Which of the following best describes the anatomical location of the incised structure?

- A. Between the cricoid cartilage and trachea
- B. Between the epiglottis and arytenoid cartilages
- C. Between the hyoid bone and thyroid cartilage
- D. Between the vocal cords in the laryngeal ventricle
- E. Between the thyroid and cricoid cartilages

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SOM.MKII.BPM2.1.NB.1.ANAT.1168

Compare and contrast structures incised during a tracheotomy versus a cricothyrotomy procedure

11. A 59-year-old man comes to the clinic because of a 1-week history of difficulty initiating swallowing, accompanied by nasopharyngeal regurgitation and a sensation of food sticking in his throat. He also reports choking and coughing while eating, as well as excessive drooling. A videofluoroscopic swallowing study is performed and a diagnosis of oropharyngeal dysphagia is made. Nasal regurgitation is a characteristic symptom of this condition, with patients experiencing regurgitation of food through their nose while swallowing. Which of the following muscles is most likely normally involved in preventing this symptom?

- A. Tensor tympani
- B. Mylohyoid
- C. Levator veli palatini
- D. Geniohyoid
- E. Salpingopharyngeus

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SOM.MKII.BPM2.1.NB.1.ANAT.0123

Describe the muscles, function and innervations of the soft palate.

12. A 52-year-old man comes to the physician because of a 2-month history of difficulty speaking and moving his tongue. He has not had difficulty swallowing or sensory deficits. MRI of the brain and neck shows a mass compressing the right hypoglossal nerve along its extracranial course. Which of the following best describes the clinical consequence of this lesion?

- A. Deviation of the tongue to the contralateral side with hyperreflexia
- B. Deviation of the tongue to the ipsilateral side with fasciculations
- C. Loss of taste sensation in the anterior two-thirds of the tongue
- D. Loss of general sensation in the posterior one-third of the tongue
- E. Weakness of the pharyngeal constrictor muscles and soft palate

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SOM.MKII.BPM2.1.NB.1.ANAT.0202

Explain the consequence of extracranial lesions of the hypoglossal nerve

13. A 30-year-old woman comes to the physician because of a 3-month history of decreased taste on the anterior part of her tongue. Neurological examination shows normal sensation of the lower lip and cheek on the right. Taste sensation is impaired on the anterior two-thirds of her tongue. Imaging studies show a lipoma in her right infratemporal fossa. Sensory impulses from the affected nerve is most likely conveyed to which of the following nuclei?

- A. Solitarius
- B. Spinal trigeminal
- C. Ambiguus
- D. Inferior salivatory
- E. Mesencephalic
- F. Superior salivatory

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SOM.MK.I.BMP2.1.NB.1.ANAT.0169

Describe the locations of the nuclei of the facial nerve and their associated structures

14. A 21-year-old man is brought to the emergency department by his father because of a 1-hour history of severe pain in his throat while breathing that started after he swallowed extremely hot coffee that 'went the wrong way' and made him cough violently. Fiberoptic examination shows second degree burns in the vestibule of the larynx. Which of the following arteries most likely supplies the affected area?

- A. Inferior laryngeal branch of inferior thyroid
- B. Superior laryngeal branch of superior thyroid
- C. Anterior glandular branch of superior thyroid
- D. Posterior glandular branch of superior thyroid

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SOM.MKII.BPM2.1.NB.1.ANAT.0130

Describe the arterial supply of the larynx,
and its venous and lymphatic drainage

15. A 34-year-old man comes to the physician because of 1-month history of a painless, slow-growing left sided mass beneath his jaw. Physical examination shows the mass is located lateral to the left anterior belly of digastric muscle and superior to the left posterior belly of digastric muscle. Imaging studies show the mass invading into floor muscles. Which of the following muscles is most likely involved?

- A. Sternothyroid
- B. Superior belly of the omohyoid
- C. Thyrohyoid
- D. Mylohyoid
- E. Sternohyoid

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SOM.MKII.BPM2.1.NB.1.ANAT.0118:

List the general attachments, innervations and actions of the muscles that are found in the floor of the mouth

16. A 62-year-old man comes to the physician because of 2-month history of an enlarging lump in his neck and occasional deep-seated right ear pain. Physical examination shows a firm and enlarged submandibular lymph node on the right. Ear examination shows no abnormalities. Endoscopic evaluation of the pharynx and larynx shows the presence of squamous cell carcinoma. Assuming that the patient's ear pain is mediated by the same nerve that innervates the tumor, which of the following is the most likely site of the tumor?

- A. Anterior 2/3rd of the tongue
- B. Anterior portion of the nasal septum
- C. Mucosa of the hard palate
- D. Mucosa of the cheek
- E. Posterior 1/3rd of the tongue

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SOM.MKII.BPM2.1.NB.1.ANAT.0115

List the nerves that carry general sensation and taste sensation from the tongue